Chapter 3, Partial Solutions

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1 [Problem not from Vallis]. Use the linear shallow water equations of motion on an f-plane, (3.98), to show that a pure inertial oscillation, with frequency f_0, has zero height perturbation. Then use the equations of motion to show that it can have neither vorticity nor divergence, and thus must have a horizontally uniform velocity field, consistent with the fact that the frequency approaches f_0 as the wavelength goes to infinity.

Let $(u, v, h) \sim e^{i(kx+ly-\omega t)}$, with $\omega = f_0$. Then substitute that into (3.98a,b) to obtain

$$f_0(-iu' - v') = -ikgh \tag{1}$$

$$f_0(-iv' + u') = -ilgh. \tag{2}$$

Multiply (1) by $-i$ and add to (2) to obtain

$$-k - il = 0. \tag{3}$$

But k, l are real by construction, so (3) can only be true if k, l are both zero. Thus $\partial h/\partial x = \partial h/\partial y = 0$. In fact we have also already answered the latter part of the question too, since (3) also means the velocities are uniform and the wavelengths are infinite, which is the limit in which one obtains inertial oscillations. Just for fun another way to do it is to form the vorticity and divergence equations first, then assume the frequency is f_0, which leads to the same result.

3.8 First part essentially already done in text, leading to (3.78); just note that $h = \eta - h_b$.

(a) The new radius is twice the old radius, so the new area is four times the old. Assuming volume is conserved, the new height of the air column must be 1/4 the old volume, so we have

$$\frac{\zeta + f}{h_{old}/4} = \frac{f}{h_{old}},$$

or $\zeta = -3f/4$. Now use Stokes theorem to relate the velocity around the perimeter to the vorticity

$$\Gamma = \int\int \zeta dA = \zeta A = \oint \mathbf{u} \cdot \mathbf{dl} = 2\pi r u,$$

where Γ is the circulation. Thus

$$u = \frac{\zeta \pi r^2}{2\pi r} = \frac{\zeta r}{2} = 3fr/8,$$

where the velocity is in the anticyclonic (clockwise) direction, since $\zeta < 0$. Using the values given,

$$u = (3/8) \times 2 \times 7.29 \times 10^{-5} s^{-1} \times \sin 30^\circ \times 2 \times 10^5 m = 5.47 m\ s^{-1}.$$

(b) Using PV conservation similarly to above gives

$$\zeta = -f/4 = 2 \times 7.29 \times 10^{-5} \times \sin 60^\circ = -3.15 \times 10^{-5} s^{-1}.$$

Assuming it moves south with no change of the fluid depth, thereafter $\zeta + f$ will be conserved, so we have

$$\zeta_{30} + f_{30} = \zeta_{60} + f_{60},$$

or

$$\zeta_{30} = \zeta_{60} + f_{60} - f_{30} = 2.19 \times 10^{-5} s^{-1}.$$

3.10 (a) The first part is worked out in the book, leading to (3.150).

(b) We'll just do it in 1D to make life easier, so $K = k$, $l = 0$, $\mathbf{u} = (0, v)$. We have for the final potential energy,

$$E_{pf} = \frac{1}{2} g \eta^2 = \frac{1}{2} g \int\int \tilde{\eta_k}^2 dk = \frac{1}{2} g \int\int \frac{\tilde{\eta_k}^2(0)}{(k^2 L_d^2 + 1)^2} dk,$$

where we have used Parseval's theorem. For the kinetic energy, use geostrophic balance:

$$E_{kf} = \frac{1}{2} H v^2 = \frac{1}{2} \frac{g^2 H}{f^2} (\frac{\partial \eta}{\partial x})^2 = \frac{1}{2} g \int\int \frac{L_d^2 k^2 \tilde{\eta_k}^2(0)}{(k^2 L_d^2 + 1)^2} dk$$

Thus the loss of potential energy (expressed as a positive number) is

$$\begin{aligned} E_{pi} - E_{pf} &= \frac{1}{2} g \int\int \tilde{\eta_k}^2(0) [1 - (k^2 L_d^2 + 1)^{-2}] dk \\ &= \frac{1}{2} g \int\int \frac{k^4 L_d^4 + 2k^2 L_d^2}{(k^2 L_d^2 + 1)^2} \tilde{\eta_k}^2(0) dk, \end{aligned}$$

which is greater than the increase in kinetic energy computed above.

(c) (Interpretation of mass field adjusting to velocity at small scales, and vice versa at large scales.)

Answers given in the book, but you have to interpret it right. For the case where the initial height perturbation is in the mass field (i.e., η), the final solution is given by (3.151), as discussed above. For large scales, $K^2L_d^2 \ll 1$, this gives $\tilde{\eta}_{k,l} \approx \tilde{\eta}_{k,l}(0)$, that is, no change in the mass field. The velocity field must adjust to achieve geostrophic balance, but if the mass field hasn't changed, it makes sense to say that that controls the result. On the other hand, if we start from an initial vorticity anomaly with no corresponding height anomaly, the book gives the result (3.152), which says that

$$\tilde{\psi}_{k,l} = \frac{K^2L_d^2\tilde{\psi}_{k,l}(0)}{K^2L_d^2 + 1}.$$

Clearly this gives $\tilde{\psi}_{k,l} = \tilde{\psi}_{k,l}(0)$ in the limit as $K^2L_d^2 \to \infty$, so for *small scales* (compared to L_d) it is the flow field that is unchanged, and to which the mass field must thus adjust to maintain geostrophy.